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FEDS Notes

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The State of AI Competition in Advanced Economies

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Global competition in artificial intelligence (AI) has intensified in recent years. Some assessments emphasize US exceptionalism, while others argue that China is eroding US dominance. By contrast, the progress of other advanced foreign economies (AFEs) receives far less attention. Existing cross-country comparisons rely largely on composite indices that, while useful as benchmarks, are subject to weighting and aggregation biases that may obscure important dimensions of AI capacity. A clearer understanding of cross-country AI capabilities can help better contextualize global AI competition.

This note brings together international comparisons on key metrics to assess countries' relative preparedness and performance in AI. The analysis shows that the United States retains important advantages in infrastructure (with the exception of electricity-related infrastructure), compute capacity, and investment conditions. AFEs face greater challenges in scaling compute resources and attracting investment. China has achieved rapid gains in research output and adoption, though its position in enabling infrastructure is less clear.[1] Taken together, these comparisons suggest a global landscape in which, despite increased competition in research and applied domains, US strengths in core AI enablers remain durable.

1. Cross-Country Comparisons Using AI Indices

We start with several widely cited indices of AI capacity and preparedness to provide an initial view of the global landscape (Table 1).[2] The United States consistently ranks at or near the top, with the United Kingdom close behind and Italy lowest among G7 members.

China's ranking varies from just below the United States on some indices (e.g., GII, Stanford) to significantly lower than most AFEs on others (e.g., IMF, Oxford).

These indices are a useful starting point but have limitations. Methodological approaches, such as indicator selection, weighting, and aggregation, can potentially obscure core metrics of AI performance and penalize diverse national AI strategies (Greco et al., 2018). Still, these indices highlight key areas for comparison: physical and digital infrastructure, investment and firm dynamics, and adoption across the economy. The rest of this note delves deeper into these dimensions, focusing on key metrics to provide a clearer picture of global AI capabilities.

Table 1: Cross-Country Comparison in Selected Technology and Artificial Intelligence Indices

	WIPO GII		IMF AI Preparedness Index		Stanford Global AI Vibrancy Tool		Oxford Insights GAIRI	
Country	Score	Rank	Score	Rank	Score	Rank	Score	Rank
United States	62.4	3	0.77	3	70.06	1	87.03	1
Canada	52.9	14	0.71	18	15.71	14	78.18	6
France	55.4	12	0.7	22	22.54	6	79.36	4
Germany	58.1	9	0.75	9	18.49	8	76.9	8
Italy	45.3	26	0.62	36	14.11	22	71.22	25
Japan	54.1	13	0.73	12	18.47	9	75.75	12
United Kingdom	61	5	0.73	13	27.21	3	78.88	5
China	56.3	11	0.64	30	40.17	2	72.01	23
India	38.3	39	0.49	71	25.54	4	62.81	46

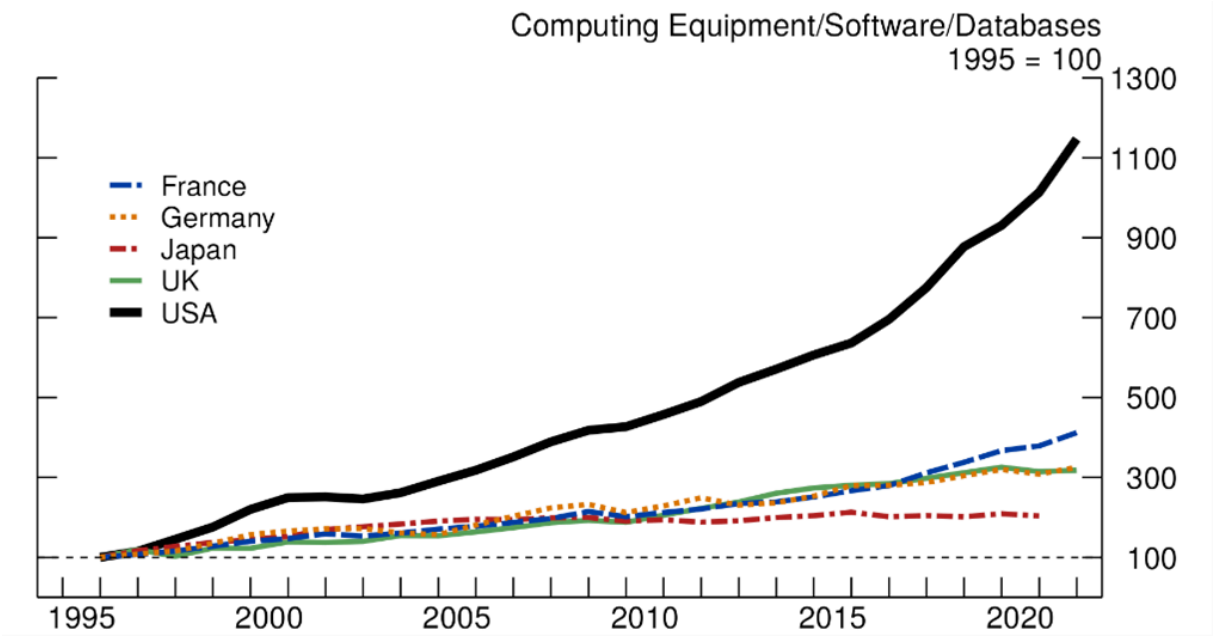
Note: WIPO GII is the Global Innovation Index published by the World Intellectual Property Organization. GAIRI refers to the Government Artificial Intelligence Readiness Index. Index data are 2025 for Stanford Vibrancy Tool, 2024 for Oxford GAIRI and WIPO GII, and 2023 for IMF Preparedness Index. The number of countries evaluated in each index are as follows: WIPO GII (133), IMF AI Preparedness Index (174), Stanford Global AI Vibrancy Tool (36), and Oxford Insights GAIRI (188).
Sources: World Intellectual Property Organization (WIPO), IMF, Stanford University, Oxford Insights.

2. Physical and Digital Infrastructure

Physical and digital infrastructure form the foundation for AI adoption. Physical infrastructure includes data centers and high-performance computing hardware, supported by reliable energy, telecommunications, and manufacturing capabilities. Digital infrastructure adds cloud computing, data access, and other tools for widespread model deployment.

As shown in Figure 1, the United States made early, outsized investments in computing, software, and databases, with annual real investment in these areas growing over tenfold from 1995 to 2021, far outpacing AFE peers, whose growth was two- to fourfold. These early investments provided the computing power, networks, and hardware that positioned the United States to lead early in AI-related innovation and diffusion.

Figure 1. Investment in AI-Related Physical and Digital Infrastructure



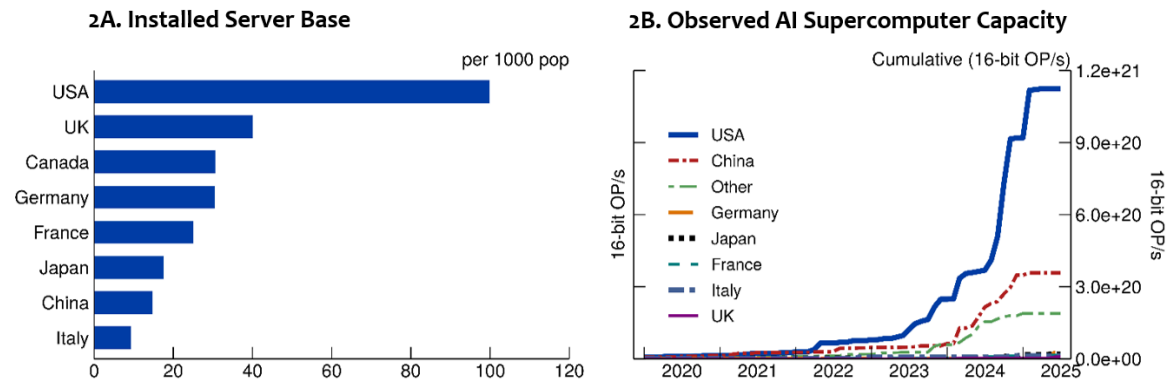
Sources: EUKLEMS , FRB Staff Calculations.

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Compute, the processing power and network resources used for AI training and inference, is the clearest measure of a country's ability to develop and deploy AI. Data centers, processors (CPUs and GPUs), and broader computing systems support the growing demands of modern AI workloads.

Data center construction in the United States surged over the past decade.[3] The United States hosted an estimated 4,049 data centers as of 2024, far more than the EU (~2,250), the UK (484), and China (379).[4] In 2024 alone, the United States added 5.8 gigawatts (GW) of data center capacity, compared with 1.6 GW in the EU and 0.2 GW in the United Kingdom.[5] On a per capita basis, the US server base stands at 99.9 per 1,000 people, far surpassing other advanced foreign economies and China (Figure 2A).

Figure 2. Compute Capacity Indicators



Notes: Right panel represents the sum of observed AI supercomputer capacity, covering an estimated 10–20% of the existing global aggregate. Data for Installed Server Base are 2024. Data for AI Supercomputer Capacity are through July 2025.
Sources: Tony Blair Institute for Global Change, Omdia, Epoch AI.
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High-end computing, particularly AI supercomputers, is particularly critical for large-scale model training and more recently inference-based tasks. The United States dominates

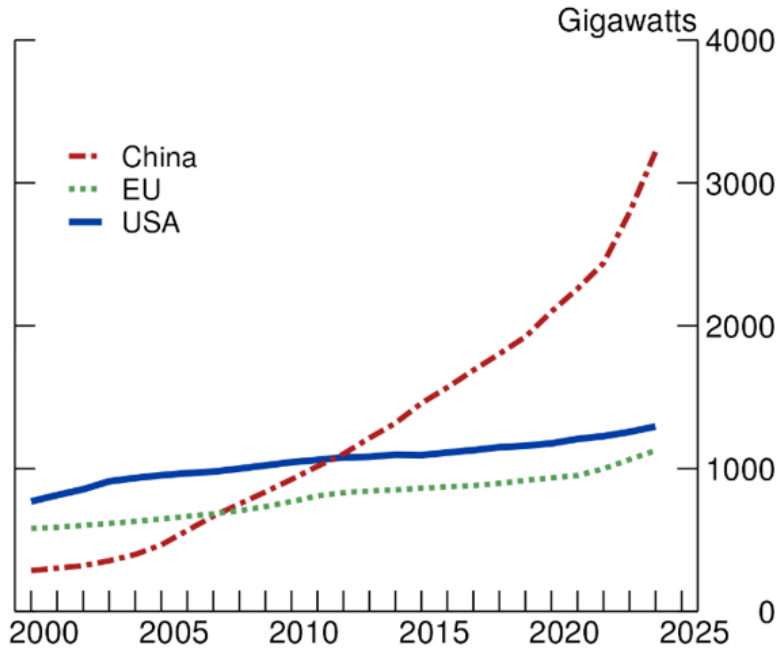
cumulative AI supercomputer capacity, controlling an estimated 74 percent of global high-end AI compute, while China holds 14 percent and the EU 4.8 percent (Figure 2B).

3. Energy Infrastructure Risks

The rapid growth of AI workloads is driving a substantial increase in data center power demand, highlighting the importance of electricity generation and transmission capacity in supporting AI infrastructure. RAND estimates that AI could require 117 GW globally by 2028, while the IEA projects that global electricity consumption from data centers will more than double by 2030, with the United States and China accounting for 80 percent of the growth.[6]

China outstripped the United States in electricity generation capacity more than a decade ago (Figure 3), with roughly 3,200 GW of installed capacity compared with 1,293 GW in the United States and 1,125 GW in the EU. China added 429 GW of net electric generation capacity in 2024 alone, over 15 times the net capacity added by the United States (Chan et al., 2025).

Figure 3. Installed Electric Generation Capacity



Source: International Energy Agency.
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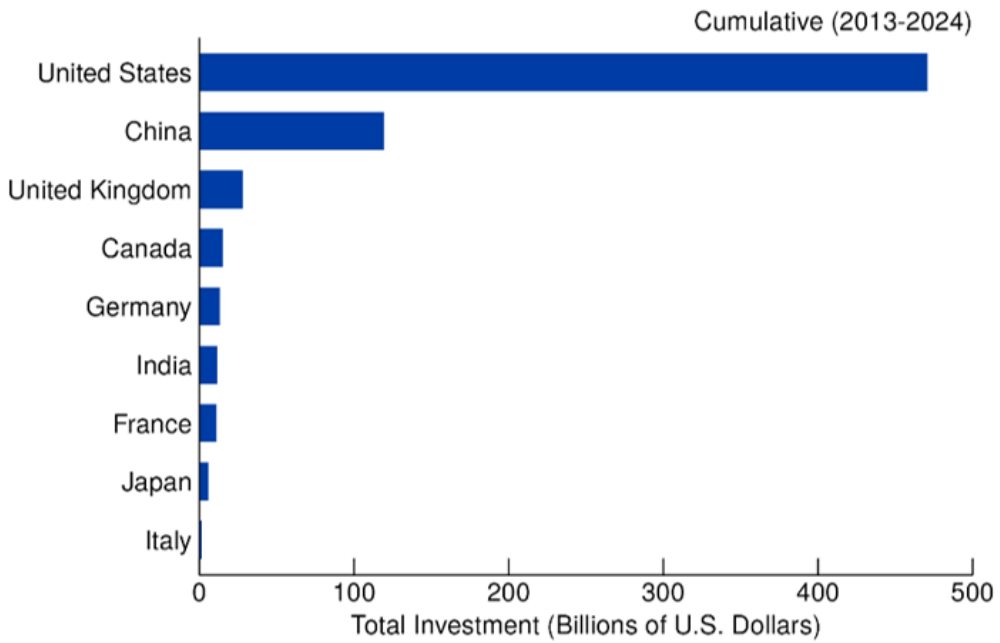
It remains unclear whether US power generation and transmission capacity will keep pace with rapidly growing data center demand. Installed data centers currently account for about 8.9 percent of average US energy use, compared with 4.8 percent in the EU and 2.3 percent in China, highlighting the scale of additional supply that may be required in the United States to support continued expansion of AI infrastructure.[7]

4. Private Investment and R&D

US private investment in AI far outpaces that of other advanced economies. Figure 4 shows that, from 2013 to 2024, cumulative private AI investment in the United States exceeded $470 billion, compared with roughly $50 billion across EU countries, $28 billion in the United Kingdom, $15 billion in Canada, and $6 billion in Japan.[8] The gaps between the United States and other advanced economies in private investment are most

pronounced in AI infrastructure, research, and data management and processing.[9] The United States also accounts for the majority of global venture capital investment in AI-related data startups and over 75 percent of reported funding in generative AI ventures.[10]

Figure 4. Private AI Investment

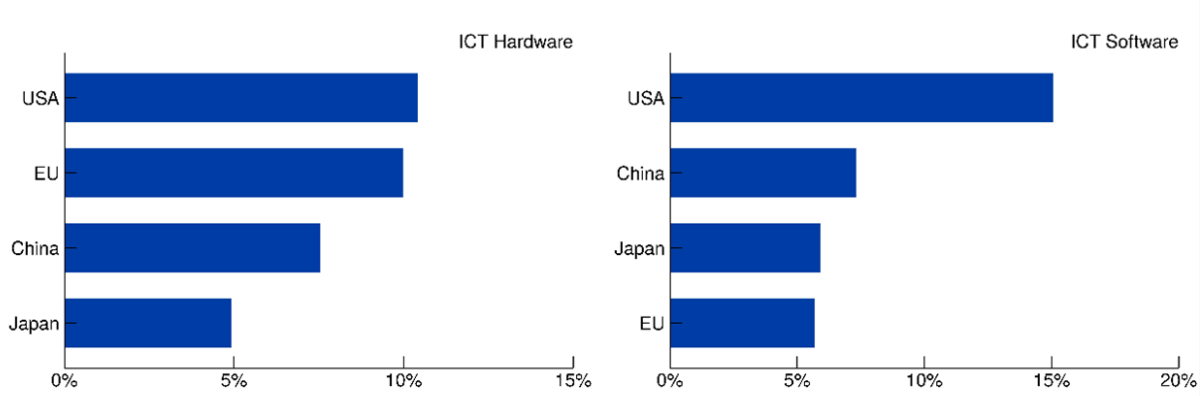


Source: Stanford 2025 AI Index Report.
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Firm-level research and development in AI-adjacent sectors further underscores US AI leadership. R&D intensity, measured as R&D spending relative to sales, is highest among US firms in both ICT hardware and software sectors (Figure 5). The EU trails slightly in hardware but falls far behind in software R&D, despite strong growth over the past decade. Chinese firms scaled hardware R&D compared to a decade prior but reduced software intensity, both of which remain far behind the United States. These patterns align

well with recent OECD findings showing that a larger share of US firms conduct AI-specific R&D for commercial use than their AFE peers (OECD 2025).[11]

Figure 5. Firm R&D Investment (% of Sales Revenue)



Notes: The EU R&D Investment Scoreboard covers the largest 2000 global companies in terms of research and development investment. Data are 2023.
Source: 2024 EU Industrial R&D Investment Scoreboard.
Accessible version

5. Domestic Adoption of AI

Comparing AI adoption across countries faces several challenges. Survey-based measures from private institutes, companies, and universities often report much higher adoption than national statistics, largely because large firms in AI-intensive sectors are overrepresented (larger firms in general have higher adoption rates across countries and surveys). It is therefore unsurprising that the United States ranks highest in surveys such as Stanford's AI Index. National statistics aim for representative samples, which include more small and medium-sized enterprises that are less likely to use AI for various reasons (see Appendix C).

Differences in definitions further complicate cross-country comparisons. For example, the US Census Bureau's biweekly BTO survey measures consistent AI use within the past

two weeks, while Eurostat reports annual adoption with less specificity on intensity. As highlighted by Crane, Green, and Soto 2025, accounting for employment weighting in these surveys substantially raises estimated US adoption, suggesting that firm-level responses alone may understate the diffusion of AI.[12]

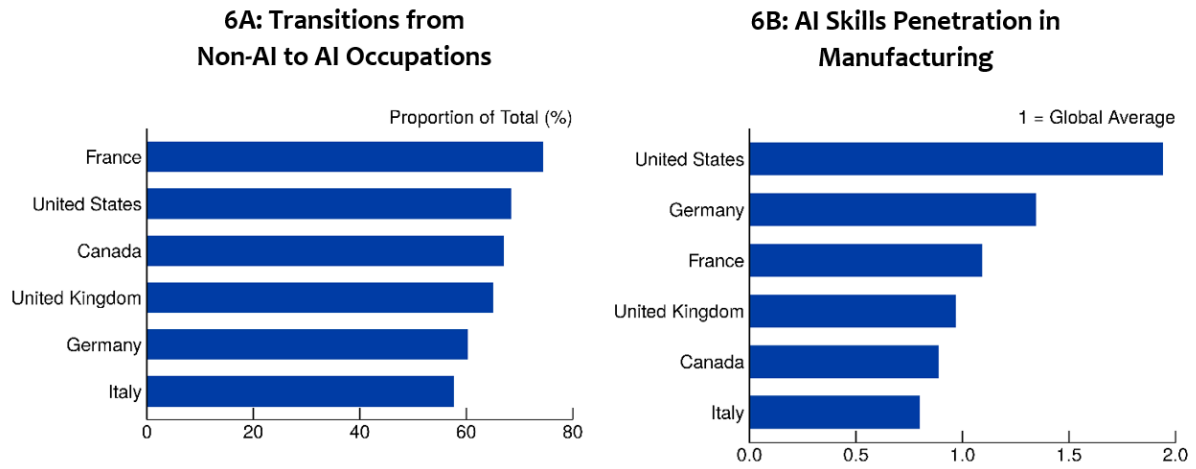
Table 2: Selected Survey Results on AI Adoption

Entity	USA	Canada	France	Germany	Italy	UK	Japan	China
National Statistics Agencies (*)	9.4%	12.2%	9.9%	19.8%	8.3%	9.0%	-	-
University of Melbourne | KPMG International	53.0%	50.0%	51.0%	51.0%	60.0%	52.0%	50.0%	89.0%
Japan Communications Ministry (GAI)	68.8%	-	-	59.2%	-	-	26.7%	81.2%
Stanford University | McKinsey and Company		82.0%				80.0%	72.0%	75.0%

Notes: Stanford University survey only reports in regional aggregates. Adoption is defined differently across surveys and results from the table should be interpreted with caution. The Japan Communications Ministry survey focuses on generative AI use, while other surveys focus on AI more broadly.
Sources: US Census Bureau, Eurostat, Statistics Canada, UK Office of National Statistics, University of Melbourne, Japan Communications Ministry, Stanford University.

Alternative diffusion indicators, such as those focused on AI-related skills in the workplace, show the United States largely outpacing AFE peers. For example, only France surpasses the United States in fostering occupation transitions from non-AI to AI employment (a proxy for upskilling and reskilling of the workforce toward AI, Figure 6a). The United States far outpaces AFE counterparts in the incorporation of AI into production processes in manufacturing (Figure 6b), as well as in education, financial, and technology service sectors (Table A3).[13] Further development and standardization of AI adoption metrics by firms and individuals will continue to clarify where advanced economies stand with regard to economy-wide AI diffusion.

Figure 6. Selected Indicators of AI Diffusion in Labor Markets



Notes: Data are the average over the 2019-2024 period. Panel 6A represents the proportion of total transitions into AI-related occupations that occur from non-AI occupations, largely reflective of upskilling or reskilling. In contrast, transitions from AI-related occupations into other AI-related occupations would be considered churn within AI occupations. Panel 6B represents self-reported use of AI skills in the manufacturing sector, where figures are scaled to the global average.
Source: OECD Artificial Intelligence Policy Observatory. Accessed September 2, 2025.
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Conclusion

This note documents how the United States largely outperforms AFEs across key areas of AI capacity, while China remains remotely competitive in some areas, with its largest advantage in energy infrastructure. The combination of robust physical and computing infrastructure, sophisticated investment climate, and emphasis on diffusing AI use across the economy will likely allow the United States to capitalize on future benefits from advances in AI, while deficiencies in one or more of these areas may limit the gains from AI in AFEs over the near term.

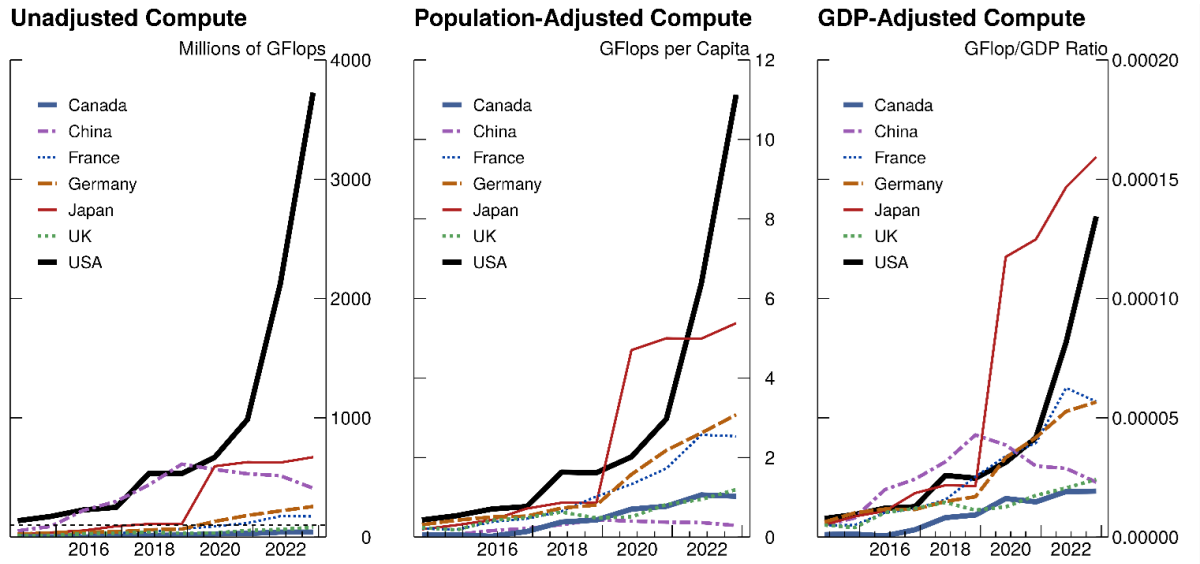
Appendix A: Additional Data

I. Compute

A common metric widely used to quantify compute capacity is Top500's supercomputer

list.[14] There are, however, multiple issues that should raise caution in interpreting Top500 results. The data represents a rotating list of the most powerful commercial supercomputers updated every six months. As such, figures across time may not reflect the same dataset of supercomputers. In addition, data suffers from self-reporting issues, particularly in the context of China, which stopped reporting its supercomputers in 2023, leading to potential overestimation for other countries on this list.

Figure A1. Compute Capacity (Top500)



Notes: Top500 data only represents the 500 most advanced commercial supercomputers, with updates occurring every six months. China stopped reporting around 2023, which may lead to overestimation of compute capacity of other economies in the future based on this data. As such, results should be interpreted with caution.
Sources: Top500, World Bank, FRB Staff Calculations.
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Table A1: Additional Compute Indicators

	Installed Servers	Data Center Planned Investment (2024-26; MW)	Cloud Service Capacity (Index)	Number of Data Centers
USA	33406915	5796	100	5381
China	21151781	504.1	35.1	449
Germany	2529076	498.1	16	521
Japan	2167381	306.6	8.8	219
India	2042480	622.6	6.1	152
UK	2709826	192.2	17.7	514
France	1620319	12	7.1	315
Canada	1209936	124.5	9.5	336

Source: Tony Blair Institute for Global Change, Cloudscene.

II. Investment

The United States has accounted for most global venture capital (VC) investment in AI and data startups and more than 75 percent of reported private VC investment in generative AI startups (Table A2). US VC investments to date were concentrated in IT, media/marketing, healthcare, and financial firms, where it maintains an outsized lead compared to both China and AFE counterparts. China maintains a slight advantage over the United States in private VC funding toward education and training startups. China also invests heavily in AI compute startups, the only country remotely close to the US level of VC investment in the sector. European and Japanese counterparts, in contrast, lag behind the United States and China across all categories by a considerable margin.

Table A2: Cumulative Venture Capital Investment in AI (Millions USD; 2012-2024)

	USA	EU27	UK	Canada	Japan	China
Total Private VC in AI Startups	598361	72781	39183	19624	11915	276380
Selected Sectors						
IT Infrastructure/Hosting	93376	6555	2203	2624	633	19168
Media/Social Platforms/Marketing	65574	4932	2014	1903	982	19640
Healthcare/Biotech	63479	4061	5874	1379	983	16994
Financial and Insurance Services	49966	11057	5399	1749	966	16285
Business Processes/Support Services	42842	10224	2853	2282	2108	13909
Education and Training	4337	496	1178	79	173	7918
Government/Defense/Security	7817	1189	-	-	-	913

	USA	EU27	UK	Canada	Japan	China
Other AI Startups						
Generative AI Startups	36435	2138	591	382	199	2843
AI Compute Startups	51866	2670	2107	3795	638	40819

Source: OECD Artificial Intelligence Policy Observatory (2025). Accessed September 2, 2025.

III. Labor Markets

Table A3: Relative AI Skills Penetration in Advanced Economies (Global Average = 1)

	USA	Canada	France	Germany	Italy	UK
Total	2.6	1.3	1.2	1.3	0.9	1.4
Selected Sectors						
Education	2.1	1.2	1.2	1.1	0.7	1.1
Financial Services	1.9	1.1	1.2	1.0	0.7	1.1
Manufacturing	1.9	0.9	1.1	1.3	0.8	1.0
Professional Services	2.5	1.1	1.2	1.3	1.0	1.2
Technology, Information, and Media	2.3	1.1	1.1	1.1	0.7	1.1

Notes: Skills frequencies are based on self-added skills by LinkedIn members in a given occupation or industry, then re-weighted using a TF-IDF model, capping at the 50 most representative skills in a given industry or occupation. The share of skills that belong to the AI skill group are then taken as a share of skills belonging to that entity. The global average of AI skills penetration is set at 1 over the 2015-2023 assessment period. For more details on the methodology, see the OECD's methodological note on its use of LinkedIn data.
Source: OECD Artificial Intelligence Policy Observatory (2025). Accessed September 9, 2025.

Appendix B: Contextualizing Competition with China

This appendix highlights important considerations when comparing AI developments between the United States and China. Comparisons between the United States and China are complicated by several factors, including a lack of transparency into Chinese data associated with AI but also a markedly different approach to AI investment and adoption across the domestic economy. We highlight only a few of these differences here.

The rise of DeepSeek, which performs at or near US model capabilities across several benchmarks, demonstrates that in terms of model development, China maintains a competitive position relative to the United States. Experts emphasize, however, that these

model capabilities are largely isolated to China's model training capabilities.[15] This model training tells little of the broader AI ecosystem, composed of compute infrastructure, investment, and adoption across the economy that facilitate widespread AI diffusion.

Data availability on Chinese compute capacity is opaque, partly resulting from ongoing geopolitical tensions and export restraints on high-end computing materials. In response to these restraints, China has focused on a combination of circumvention and building domestic capacity, both of which it keeps largely veiled. As such, key indicators of China's position relative to the United States in AI, such as high-end compute capacity, are unreliable.[16] Widespread smuggling of high-end chips and other circumvention of US export controls may also result in underestimating China's overall compute capacity.[17]

Some metrics, such as private investment, do not adequately capture the nature of AI investment in the Chinese economy, which focuses on a state-driven approach with local governments as diffusing agents. For example, Stanford researchers recently reported that over the past decade, China plowed $912 billion of investment into several key sectors through vehicles known as local government venture capital funds, with roughly $184 billion going to nearly 10,000 AI-related firms across China (Beraja et al., 2024; Omaar 2024). These local government VC funds tend to invest in a broader geographical profile across China compared to private VCs (which largely cluster in coastal regions), which may result in several benefits including more widespread dispersion of AI across the entire Chinese economy. In March 2025, China announced an additional venture capital guidance fund dedicating ~$138 billion over 20 years to AI and quantum technology.[18] These funds, however, could likewise generate poor innovation returns on investment if funds are poorly allocated across firms due to, among other things, local government corruption.

Figures on China's AI infrastructure buildout may not fully capture inefficiencies that currently limit China's ability to close to the gap with the United States. For example, local outlets as recently as March 2025 reported that up to 80 percent of China's newly built computing resources were idle, as many companies running the data centers struggled to stay afloat.[19] A misallocation of resources by both local officials and corporations seeking investment alternatives during the real estate slump reportedly led to the construction of

data centers that fell short of industry standards and suitability requirements for ongoing AI market developments (e.g., the shift from pre-training workloads to inference-based tasks). This shift to inference and real-time reasoning, particularly in the advent of DeepSeek, reinforced the need for lower latency, which disadvantages Chinese data centers recently built in rural areas that benefit from cheap land and energy prices but face latency issues. In response to this data center glut, the Chinese government is currently conducting an assessment to regulate and optimize existing resources and sell off unused compute power (Reuters 2025). It remains unclear at present whether China can successfully repurpose these facilities for productive means, AI or otherwise.

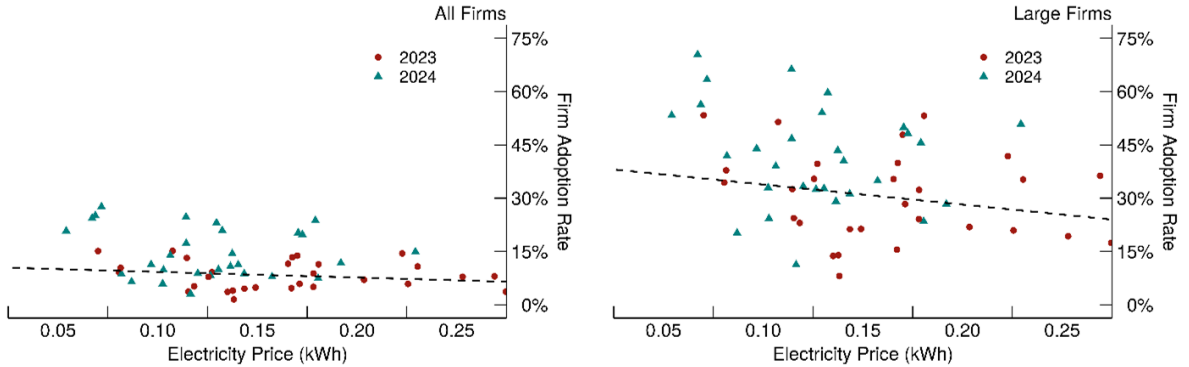
Appendix C: Reasons for Lagging Adoption among Other AFEs

Drawing from a range of country-specific reports, this appendix reviews the main barriers that help explain why AFEs lag the United States in AI adoption. Some factors repeat earlier themes, such as limited computing power, while others relate to costs, skills, and finance.

- **Lack of Computing Power Infrastructure**: As discussed in Section 2, many AFEs face a shortage of computing capacity (Dobbs and Hirsch-Allen 2024; Schnabel 2024; Li 2024). The Euro Area missed out on earlier buildouts of IT and data infrastructure, while Canada reports gaps in AI-specific infrastructure. The United States has more than four times the compute performance capacity of its closest G7 peer, though this lead narrows when adjusting for GDP or population. In Japan and the United Kingdom, infrastructure exists but is not broadly accessible, especially for small and medium sized enterprises who struggle to compete for compute access with larger firms.
- **Skills and Training Gaps**: Most AFEs report shortages of workers with AI expertise (Canadian Chamber of Commerce 2024; Oliveira-Cunha et al., 2024; Hoffman and Nurski 2021). Firms also struggle to retrain existing employees for AI tasks, indicating a general mismatch in skills. As previously shown in Figure 6, for some sectors like manufacturing, AI skill penetration in several G7 economies is at or below the global average.

- **Access to Equity and External Investment**: In the United Kingdom and Euro Area, many firms cite limited access to equity investment as a barrier (European Commission 2025). The issue is particularly acute for SMEs in Europe, which cite high upfront costs to AI implementation and more binding credit constraints than larger firms (Hoffman and Nurski 2021). In the European case, one avenue to alleviate these constraints could be the deepening of the internal market and further progress on the savings and investment union (Airaudo et al. 2025).
- **Energy costs**: High electricity prices, particularly in Europe, might constrain adoption. As shown in Figure C1, country level data from the past two years show a slight negative correlation between energy costs and AI use among firms, with the effect significantly stronger for large enterprises. This trend is complicated by the fact that increased AI infrastructure buildout is causing a surge in electricity demand, putting upward pressure on prices.

Figure C1. Electricity Price and AI Adoption



Notes: Electricity data are for the first six months of each year. Large firms specifies those with 250 or more employees. Fit lines are based on a simple bivariate regression with year fixed effects.
Source: Eurostat, FRB Staff Calculations.
Accessible version

Table Appendix D: Brief Literature Review of AI's Effects on Productivity and Employment

Authors	Method	Country	Year(s)	Findings
Acemoglu 2024		United States		Modest TFP gains (upper bound of 0.66% over 10-year span). GDP effects of between 0.93%-1.16% over next 10 years. No substantial negative wage effects.
Acemoglu et al., 2022		United States	2010-2018	Reduction in hiring in non-AI positions and change skill requirements of remaining postings (at the establishment level; no detection of such effects at sectoral or occupational levels).
Aghion et al., 2019		France	1994-2014	Effect of robotization reduces aggregate employment and at the employment zone level, with greater declines for non-educated workers.
Babina et al., 2024	FE Model	United States	2007; 2010-2018	AI-investing firms experience higher employment and market valuations, relative growth among larger adopting firms, and increased industry concentration.
Bonfiglioli et al., 2024	Shift-Share IV	United States	2000-2020	AI exposure negatively associated with employment across commuting zones and time. Employment effect lower for low-skill and production workers, positive for workers at higher end of income distribution and STEM.
Czarnitzki et al., 2023	Panel IV	Germany	2018	Positive association between AI adoption and firm productivity.
Damioli et al., 2021		Global	2000-2016	AI patent applications associated with positive effect on company's labor productivity, concentrated in SMEs and service industries.
Eloundou et al., 2023		United States		Findings suggest 80% of US workforce could have at least 10% of work tasks affected by LLM introduction (19% of workers with at least 50% of tasks).
Filippucci et al. 2024 (OECD)	Surveys	9 OECD Economies	2016-2021	AI use associated with higher productivity, but insignificant marginal effect when controlling for use of other ICT.
Filipucci et al., 2025 (OECD)		G7	2022-2023	Labor productivity gains from AI expected at between 0.2 and 1.3 percentage points over the next decade, depending on AI adoption speed, and with significant cross-country heterogeneity due to, among other things, sectoral composition.
Hoffman et al., 2024	Regression Discontinuity	United States	2022-2023	Generative AI access reorients skills towards more productive uses, with greater effects for lower-skilled workers, implying a potential equalizing effect on productivity gaps.
Shen and Zhang 2024	Two-Way Fixed Effect Model;	China	2006-2020	Introduction of AI technology increases number of jobs, with greater job-share increases for women an workers in labor-

Authors	Method	Country	Year(s)	Findings
	2SLS			intensive industries.
Yang 2022	GMM; Dynamic Panel Data Model	Taiwan	2002-2018	Significant, positive association between AI technology and firm productivity and labor demand.

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1. We discuss the nuances of China's infrastructure position relative to the United States in greater detail in Appendix B. For more on China's research and innovation capacity, see Ates and Jeon 2025. Return to text
2. Each index varies in its measurement of AI-related capacity. The WIPO Global Innovation Index (GII) focuses on metrics aimed at assessing innovation capacity more broadly. Stanford's Global AI Vibrancy Tool focuses on eight pillars of AI-related activity, including research and development, infrastructure, education, governance, and public opinion. Oxford's Global AI Readiness Index (GAIRI) emphasizes AI buildout and use as it pertains to the public sector and governance more generally. The IMF's AI Preparedness Index evaluates

countries on four dimensions focused on digital infrastructure, human capital, innovation, and regulatory frameworks. Return to text

3. $41.2 billion (annualized) was put in place in July 2025. US Census Bureau. Data are seasonally adjusted. Return to text

4. Estimates are based on data from Data Center Map (for alternative Cloudscene reporting, see Appendix A). Data center estimates vary greatly across studies, but all consistently place the United States far beyond its AFE counterparts and China. Return to text

5. State of Global Compute 2024 Report. Tony Blair Institute for Global Change. Return to text

6. In the US case, electricity demand by data centers is expected to account for half of electricity demand growth through 2030. Return to text

7. Pilz et al. (2025) provide a detailed discussion of regulatory, permitting, and infrastructure factors that influence the expansion of AI data centers. Return to text

8. For other breakdowns, see Stanford's 2025 AI Index Report. Return to text

9. Based on previous Stanford AI Index Report from 2024, which disaggregated by sector and country. Return to text

10. See Appendix A for more detail on venture capital investment across countries and sectors. Return to text

11. In the OECD's study, the authors find that 86 percent of US firms perform some degree of AI-specific R&D, compared to 75 percent in Germany, which is the next highest among G7 economies. Return to text

12. For a more detailed list of survey results of firms and employees on AI adoption in the United States and Europe, see Crane, Green, and Soto 2025. Return to text

13. For a full sectoral breakdown of AI skills penetration, see https://oecd.ai/en/data and Appendix Table A3. The methodological note on the calculation of AI-related labor market indicators based on LinkedIn data is available at https://oecd.ai/en/linkedin . Return to text

14. See, for example, https://oecd.ai/en/wonk/canadas-ai-compute-gap . Return to text

15. For a more detailed explanation, see Heim 2025. Return to text

16. For example, China stopped reporting its high-end compute capacity (e.g., supercomputers) to global indices like Top500, seen largely as a response to export controls imposed by the United States on high-performance chips and other technologies. As such, its high-end compute capacity is largely shrouded since 2023. Return to text

17. The Financial Times (https://www.ft.com/content/6f806f6e-61c1-4b8d-9694-90d7328a7b54) reported, for example, that between April and July 2025, more than $1 billion in AI processors, including Nvidia's advanced B200 chip, were shipped to China. Return to text

18. The following article from Quantum Insider provides more detail on the expected venture capital fund. https://thequantuminsider.com/2025/03/07/china-launches-138-billion-government-backed-venture-fund-includes-quantum-startups/ Return to text

19. For more detail, see Chen, Caiwei. 2025. "China built hundreds of AI data centers to catch the AI boom. Now many stand unused." MIT Technology Review. March 26, https://www.technologyreview.com/2025/03/26/1113802/china-ai-data-centers-unused/. Return to text

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